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The Impact of Fault-Fracture Interaction on Fluid Flow in Carbonates: Laboratory Experiments

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Abstract

Understanding the impact of faults and natural fractures on fluid flow in petroleum reservoirs is important at all stages of a field's development. Sealing faults compartmentalise reservoirs, reducing the rate of oil production by locally reducing permeability; the presence of sealing faults usually requires more wells or more complex wells to be drilled to achieve production targets (Smith, 1966; Berg, 1975; Watts, 1987; Alexander and Handschy, 1998; Fisher et al., 1998; Fisher et al., 2001; Fisher et al., 2005; Fisher et al., 2013). Open fractures and/or dilatant faults may enhance productivity (Coney et al., 1993) but may also cause production problems, such as early water breakthrough in producer wells (Rawnsley and Wei, 2001). Single extensional fractures may develop vuggy porosity or become partially sealed, i.e. their permeability may become modified by dissolution or karstification (Nelson 2001).

Some reservoirs can contain both open fractures and low-permeability fault rocks (Nelson, 2001; Geraud et al., 2006; Micarelli et al., 2006; Ran et al., 2014, Al Hashemi et al., 2023; Al Hashemi et al., 2024). There exists a lack of understanding of how open fractures and low-permeability faults interact and affect fluid flow. To fill this knowledge gap, a series of fluid flow experiments were conducted on carbonates with open fractures and low-permeability barriers (i.e. analogous to low-permeability faults). The main objective of the experimental study described in this paper is to assess the impact that low-permeability faults in conjunction with open fractures have on oil production. In particular, the experiments compare the production behaviours of samples with sealing faults to those in which open fractures breach the sealing faults; the latter are referred to as partially sealing faults. It is critical to emphasise that the flow behaviour achieved by core flooding in the laboratory is on a much smaller scale than that of a reservoir, so up-scaling is an issue.

Materials

Rock samples and fluids

Three carbonate core plugs with a diameter of 3.8 cm and a length of around 5.5 cm taken from different locations were used in the experiments (see Figure 1):



Figure 1—A picture of the three core plugs (Ma-22, Ma-31 and TAD-1) after cutting and reassembly.

Sample Ma-22 is an Upper Coralline Limestone sample from Malta. It is a shallow-water limestone that varies from coralline algal biostrome facies to patch reefs and platform and slope facies (Bosence and Pedley, 1982). The sediments experienced limited diagenetic alteration due to their shallow burial depth of 300 m to 1000 m (Dart et al., 1993; Kim et al., 2003; Haines et al., 2016).

Sample Ma-31 is from a Globigerina Limestone from Malta. It is represented by a succession of biomicritic, yellow-cream pelagic limestones dominated by Globigerina foraminifera. It reaches a thickness of >200 m (Pedley, 1978) and is divided into three units (lower, middle and upper), which are each underlain by thin hardgrounds and phosphoritic conglomerate layers (Pedley and Bennett, 1985).

Sample TAD-1, which is from the Cadeby Formation and is known as Magnesian Limestone (Smith et al., 1986), was collected from Tadcaster in the United Kingdom. The formation's thickness varies considerably, ranging from 15 m above Carboniferous sandstone hills to approximately 80 m in the district's east. The formation is composed mainly of dolomite but has a complicated diagenetic history and contains vugs mineralised with galena, haematite and copper minerals (Marshall, 1856; Harwood and Smith, 1986).

Formation brine with a density of 1.1 g/cm³ was used to saturate samples Ma-22 and Ma-31, while tap water with a density of 1 g/cm³ was used to saturate sample TAD-1. Mineral oil with a density of 0.76 g/cm³ was used in all samples, and the oil was dyed using Waxoline Red to make it easier to measure the volume expelled from the sample during experiments.

Sample preparation

The creation of open fractures

Naturally fractured specimens are difficult to collect from outcrops, and they have such variable permeability that it is difficult to compare experimental results from a range of samples. It was therefore decided to create artificial fractures parallel to the axis of core plugs using a diamond saw. Spacers with a thickness of 1.5 mm were placed between the surfaces to keep the fracture open. The spacers also helped to ensure the diameter of the plug remained the same by compensating for material lost during cutting. The perimeter of the plug was then sealed using a heat shrink plastic sleeve (see Figure 2).

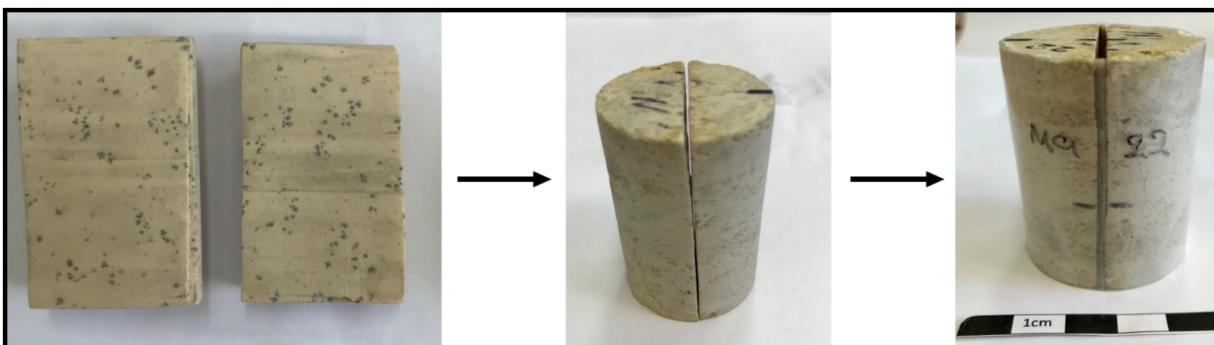


Figure 2—An example showing the steps involved in cutting and assembling the core plug Ma-22 before the fluid flow test.

The location of artificial fractures and the size of the matrix for all the plugs are shown in Figure 3.

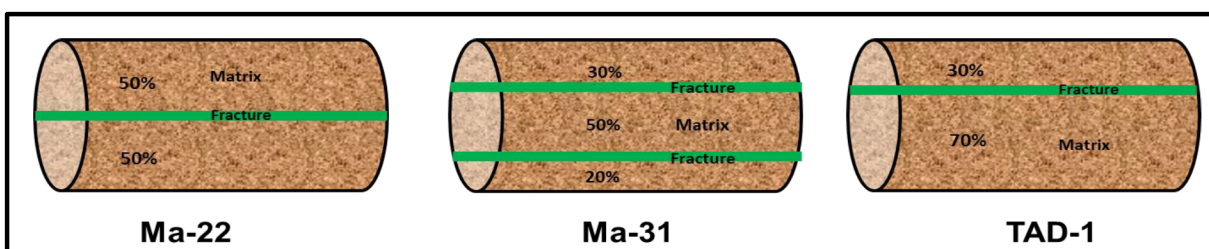


Figure 3—An illustration showing the size of each part of the three core plugs (Ma-22, Ma-31 and TAD-1).

Sample Ma-22: one fracture in the middle was simulated by cutting the core plug into two parts of equal size (50% and 50%).

Sample Ma-31: two fractures were simulated by cutting the core plug into three different sizes (30%, 50% and 20%).

Sample TAD-1: one fracture was created to divide the core plug into two different sizes (30% and 70%).

Partially sealing faults

To represent a partially sealing fault during the experiment, an impermeable disk with a 1 mm leakage hole was placed downstream of all core plugs in a location that ensured that it did not intersect directly with the fractures (see Figure 4).

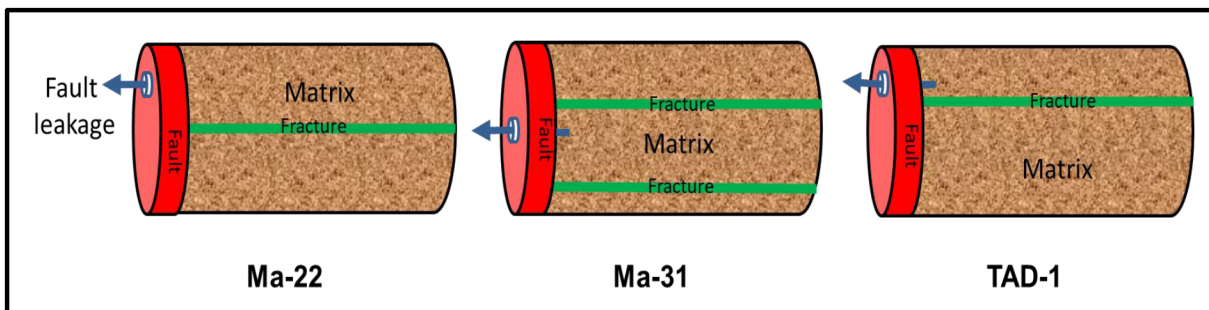


Figure 4—An illustration showing the location of the fault leakage in the three core plugs (Ma-22, Ma-31 and TAD-1).

Sample Ma-22: the fault leakage was positioned on the top part and next to the matrix to avoid direct production from the fracture. This geometry was used to examine oil production from two equally sized compartments – one sealed and the other partially sealed by the impermeable disk.

Sample Ma-31: the fault leakage was positioned on the middle part and next to the matrix to avoid direct production from the two fractures. This geometry was used to examine oil production from three parts of various sizes (upper 30%, middle 50% and lower 20%).

Sample TAD-1: the fault leakage was positioned on the top part (small part) and next to the matrix to avoid direct production from the fracture. This geometry was used to examine oil production from two parts of different sizes (upper 30% and lower 70%).

Methods

The methods section begins with a brief description of the experimental setup and procedure. Details are then provided of CT imaging and quantification (CT porosity and saturation), porosity measurement using the saturation method, pump calibration, liquid permeability (steady state), irreducible water saturation, spontaneous brine and oil imbibition tests, new fluid identification methodology (the digital multimeter method [DMM]), water flooding in samples with open fractures and water flooding in the samples with a partially sealing fault.

Experimental setup and procedure

The sample sizes were made with precision; the length and diameter of the cylindrical core plug samples were determined using a digital calliper with a resolution of 0.01 mm, and multiple measurements in each dimension (radius and height) were averaged to reduce errors. The core plugs were oven dried at 60°C for 48 hours and weighed and scanned using X-ray CT. The samples were then saturated with brine and rescanned. Porosity measurements were made using the results from the weight of the dry and saturated samples and the CT scans. The liquid permeability of the matrix was measured using steady-state flow before creating the artificial fractures.

The saturated samples were then placed in an ultrahigh-speed centrifuge to achieve irreducible water saturation, S_{wi} . The core plugs were then saturated in oil under a vacuum for at least 24 hours. A spontaneous water imbibition test was then conducted on all samples to determine the volume of oil produced before the water flooding experiments. This was followed by spontaneous oil imbibition, forced brine imbibition and forced oil imbibition tests to determine the wettability index (WI) for all the samples; the methodologies are described in detail below.

After the spontaneous brine and oil imbibition tests, the centrifuge was used to force brine out of the core plugs. The samples were submerged in oil in a container and then placed in a vacuum for at least 10 hours to allow complete oil saturation. The samples were then ready for the flow experiments. The samples were scanned with the CT scanner to determine the oil and brine saturation.

The first experiment was the water flood on the samples containing the open fractures (i.e. not partially sealing faults). After this experiment, the samples were cleaned and then re-saturated with oil to S_{wi} , using the method described above. The porous plate, to represent the partially sealing fault, was placed on the samples before water flooding.

The overall workflow for the experiments is presented in [Figure 5](#), and each stage is described in detail below.

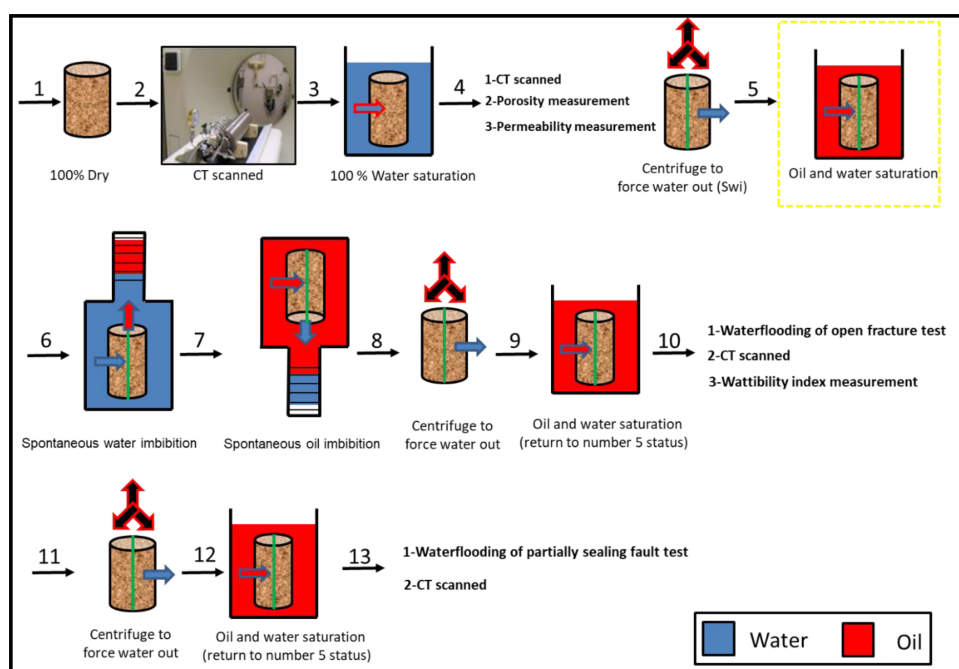


Figure 5—Illustration showing the experimental sequence for all core plugs.

Oil and water flow determination: a new fluid identification methodology

It is essential to obtain information on the production rates of oil and brine during the flow experiments. The Wolfson laboratory does not have sophisticated phase separators that can be used to automatically measure the instantaneous rate of oil and brine produced during two-phase flow experiments. The DMM was therefore developed to measure the instantaneous water and oil effluent from a core plug utilising the conductivity of the fluids produced and measuring the voltage in a four-electrode configuration. Essentially, the DMM (see Figure 6) is an instrument that consists of an AC power source, measuring probes, a Keithley multimeter and datalogging software (LabVIEW).

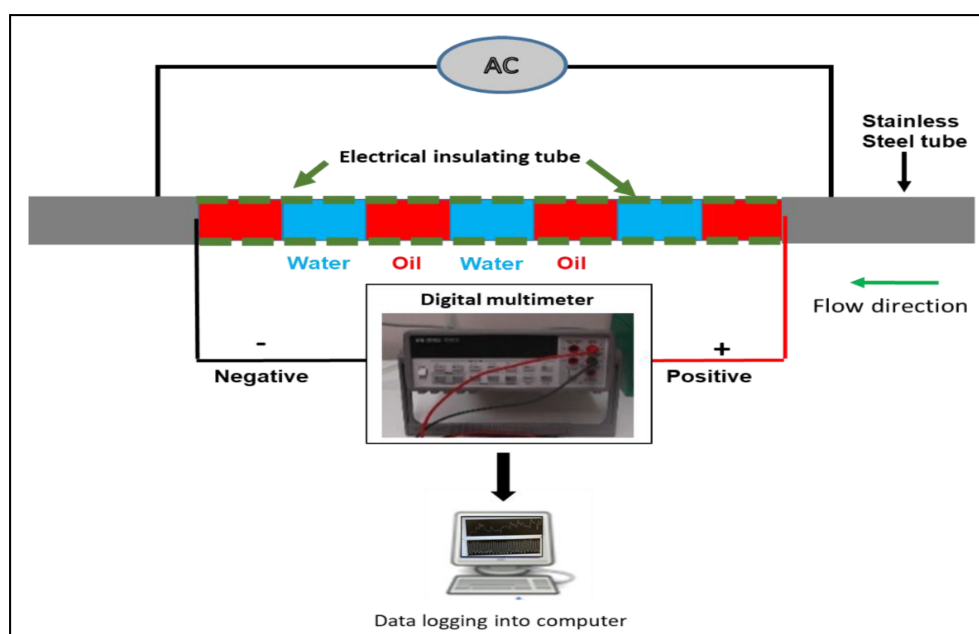


Figure 6—A schematic of the digital multimeter method.

After analysing voltage data from several flow experiments, a methodology was developed using the DMM to determine the instantaneous fluid production at the outlet of the core plug (oil or water and their rate). As a result, the sum of the volume flowing between the probes matches the cumulative fluid production collected in test tubes, as shown in Figure 7 and Figure 8. Thus, the oil to water ratio can easily be obtained as a function of time.

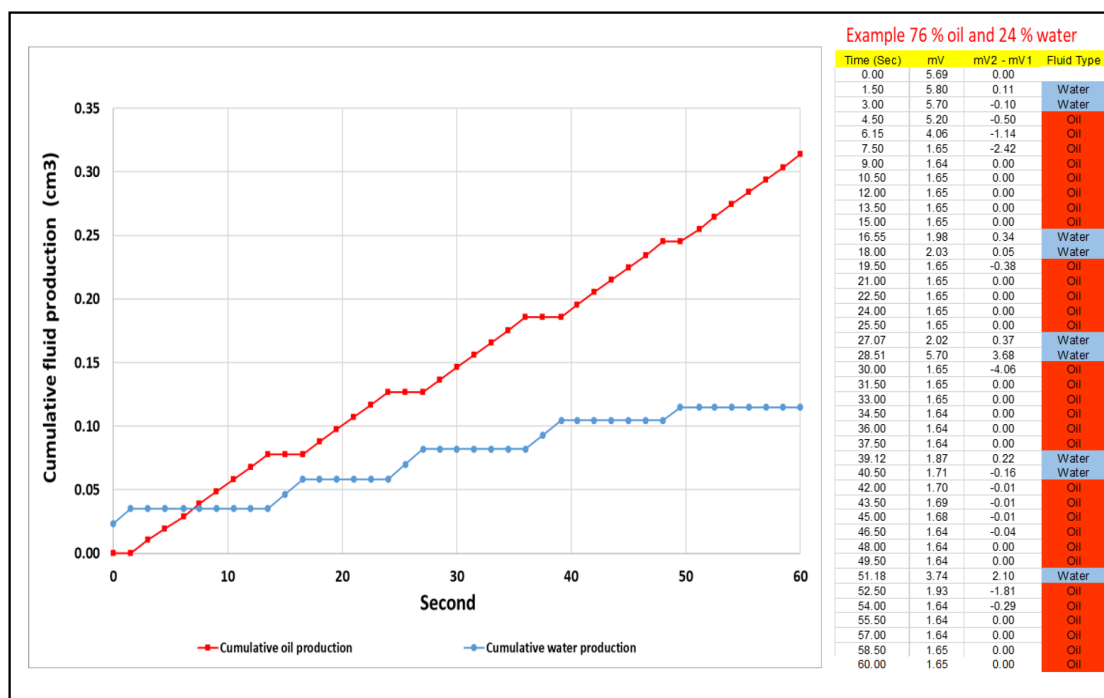


Figure 7—Example of the cumulative oil and water production calculated using the digital multimeter method for an injection of 76% oil and 24% water. The table on the right shows the successive slugs of oil and water. Blue represents water, and red represents oil.

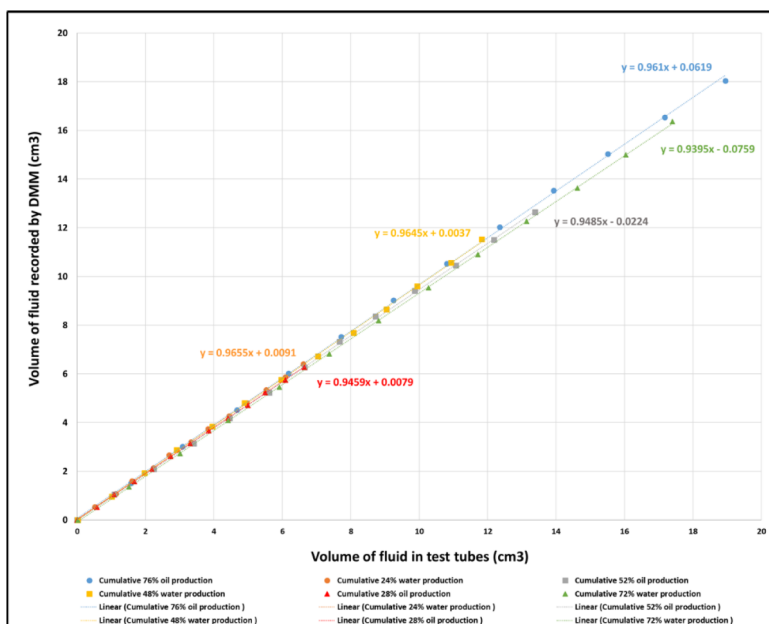


Figure 8—Verification of the digital multimeter method (DMM) for different water to oil ratios. The horizontal axis shows the volume measured in test tubes, and the vertical axis shows the volume calculated using the DMM.

Water-flooding samples with open fractures

In these experiments, the core plugs at S_{wi} were loaded in the core holder under 1,000 psi confining pressure (see Figure 9). A brine containing 8% sodium bromide, with a CT number of -249 CT units, was used in this experiment to differentiate the solution that invaded the core plugs from the brine within the core. The brine was injected at ambient conditions and at a constant rate using a Millipore Waters Model 510 pump.

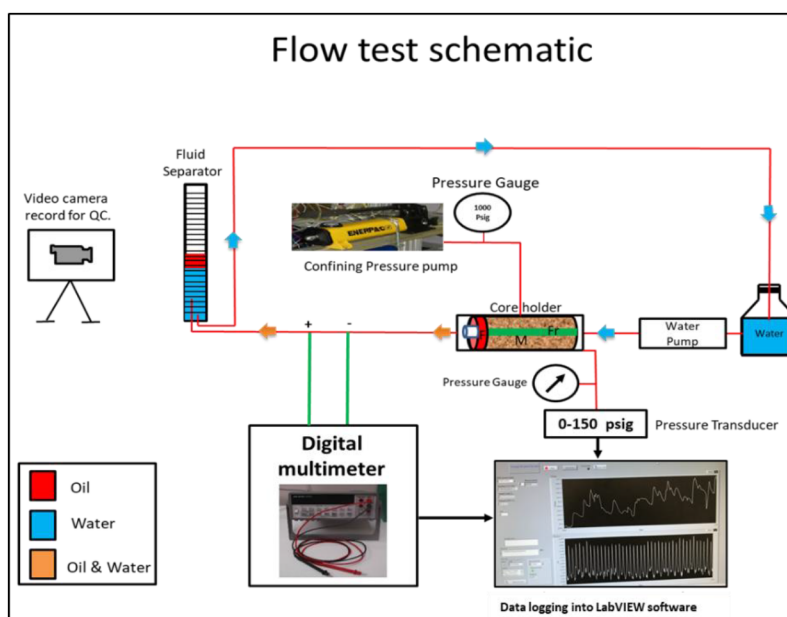


Figure 9—Diagrammatic representation of the experimental setup used during water flooding across fractured and partially sealing fault samples.

The brine was recirculated through the fractured core plugs at a constant flow rate of 0.4 ml/min. Each experiment was run until no additional oil was produced and pressure reached a steady state; the pressure and DDM voltage were recorded every 1.5 seconds using LabVIEW software. A video camera was used to record the cumulative volume of oil collected in the fluid separator to validate the production rate determined by DMM in the time function.

Results

Porosity

The porosities determined from the weight of dry and saturated samples ranged from 18.6 to 28.2% (see Table 1).

Table 1—Saturation porosity results with general information for the three core plug measurements from different outcrops and locations.

Sample number	Sample saturated	Length (cm)	Diameter (cm)	Area (cm ²)	Weight – dry (g)	Weight – wet (g)	Pore volume (cm ³)	Sat. porosity (%)
Ma-22	Brine	5.35	3.74	10.99	128.494	139.43	10.936	18.6
Ma-31	Brine	5.64	3.69	10.67	113.178	130.202	17.024	28.2
TAD-1	Tap water	5.37	3.68	10.62	126.829	138.215	11.386	19.9

Permeability

The liquid permeability of the three samples ranged from 0.56 to 6.58 mD at 1,000 psi confining pressure (see Table 2).

Table 2—Permeability results for the three core plugs.

Sample number	Sample saturated	Permeability (mD)
Ma-22	Formation brine	0.56
Ma-31	Formation brine	6.58
TAD-1	Tap water	1.50

Water flooding of the open fracture

The fracture-only core plugs were used in the first water-flooding test; the results are presented in Table 3. Sample Ma-22, which had one fracture in the middle of the core, produced 2.2 cm³ by water flooding and 2.3 cm³ during spontaneous imbibition (see Figure 10). In other words, the water flooding produced 95.7% of the oil produced by spontaneous imbibition.

Table 3—Comparison between the Amott cell test and water-flooding test for fractured core plugs.

Sample number	Oil production (cm3) from Amott cell	Oil production (cm3) from flooding test	Oil production (%)
Ma-22 (One fracture)	2.3	2.2	95.7
Ma-31 (Two fractures)	3	2.5	83.3
TAD-1 (One fracture)	0.9	0.7	77.8

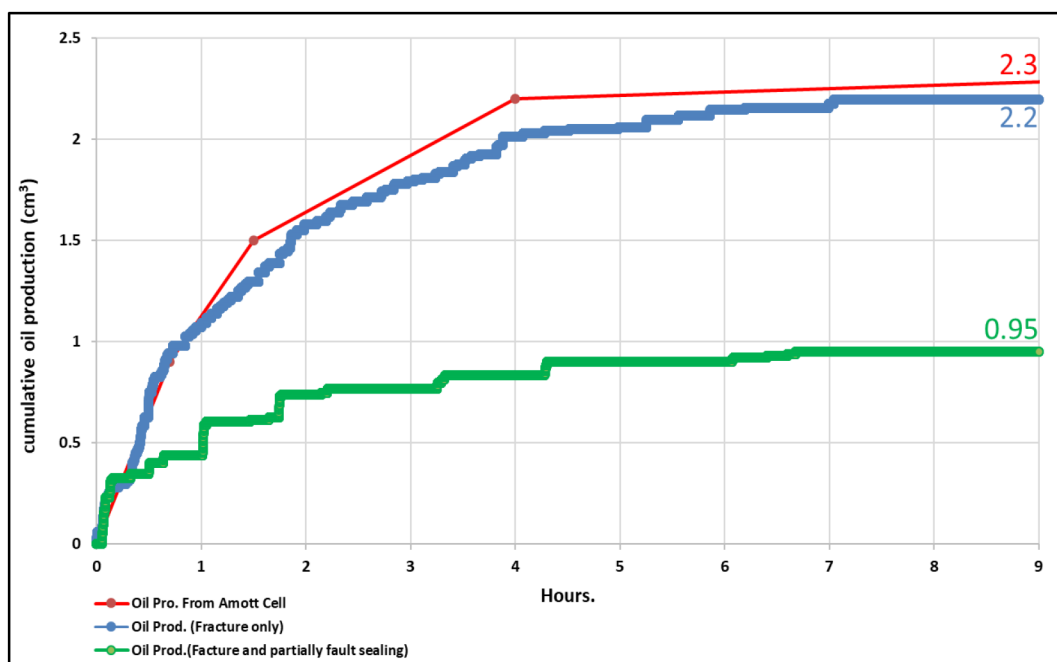


Figure 10—Cumulative oil production vs time from the tests for sample Ma-22. The red line shows oil production during the spontaneous imbibition experiment. The blue line shows oil production during the water-flooding test from the core plug containing the fracture. The green line shows oil production during water flooding on the core plug containing the fracture and the partially sealing fault.

Sample Ma-31, which contained two fractures, splitting the core plug into three parts, produced 2.5 cm³ by water flooding and 3 cm³ by spontaneous imbibition (see Figure 11). In other words, the water flooding produced only 83.3% of the oil produced during spontaneous imbibition.

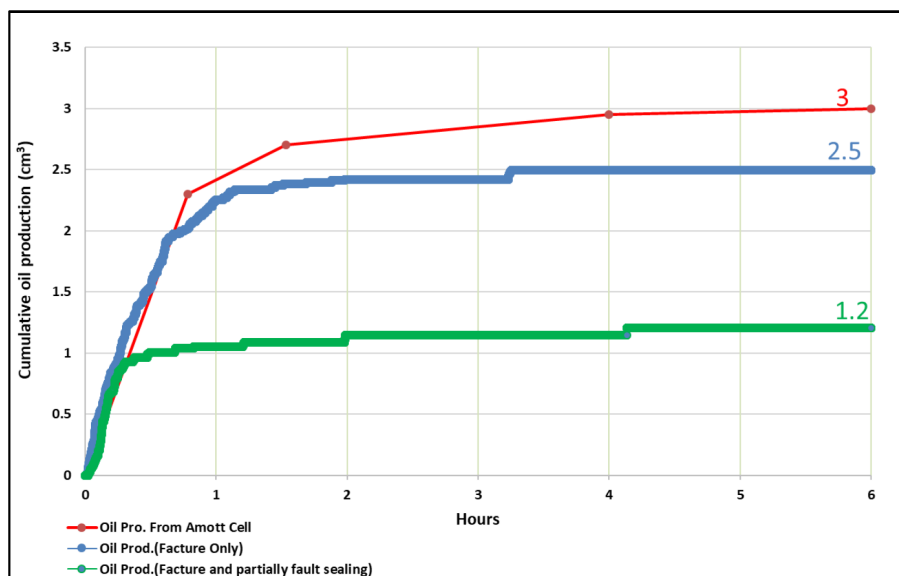


Figure 11—Cumulative oil production vs time function from the tests for sample Ma-31. The red line shows oil production during the spontaneous imbibition experiment. The blue line shows oil production during the water-flooding test from the core plug containing the fracture. The green line shows oil production during water flooding on the core plug containing the fracture and the partially sealing fault.

Sample TAD-1, which had one fracture dividing the core plug into two unequally sized pieces, produced 0.7 cm³ by water flooding and 0.9 cm³ by spontaneous imbibition (see Figure 12). In other words, the water flooding produced 77.8% of the oil produced during spontaneous imbibition.

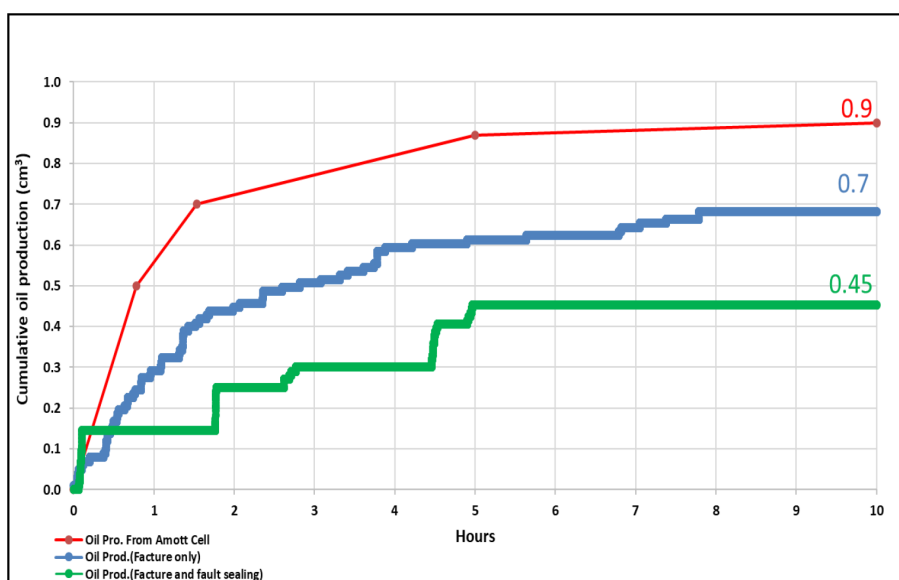


Figure 12—Cumulative oil production vs time from the tests for sample TAD-1. The red line shows oil production during the spontaneous imbibition experiment. The blue line shows oil production during the water-flooding test from the core plug containing the fracture. The green line shows oil production during water flooding on the core plug containing the fracture and the partially sealing fault.

Water flooding of the partially sealing fault

The core plugs with fractures and partially sealing faults were used for the second water-flooding test performed using the same experimental configuration and flow rates as the first test. All samples were returned to their base fluid saturation condition and readiness for the second test. The results of water-flooding core plugs with partially sealing faults used in the second water-flooding test are presented in Table 4. Sample Ma-22 produced 0.95 cm³ by the water flooding of the partially sealing fault (WPSF) test compared to 2.3 cm³ produced during imbibition (see Figure 10). In other words, water flooding produced only 41.3% of the oil produced during spontaneous imbibition. Sample Ma-31 produced 1.2 cm³ by the WPSF test compared to 3 cm³ during imbibition (see Figure 11). In other words, water flooding produced only 40% of the oil produced during spontaneous imbibition. Sample TAD-1 produced 0.45 cm³ by the WPSF test compared to 0.9 cm³ during imbibition (see Figure 12). In other words, water flooding produced only 50% of the oil produced during spontaneous imbibition. Figure 13 to Figure 15 show the upstream pressure vs time from the different tests.

Table 4—Comparison between the Amott cell test and water-flooding test for the samples with the partial sealing fault and

Sample Number	Oil production (cm ³) from the Amott cell	Oil production (cm ³) from the flooding test	Oil production %
Ma-22 (one fracture)	2.3	0.95	41.3
Ma-31 (two fractures)	3	1.2	40.0
TAD-1 (one fracture)	0.9	0.45	50.0

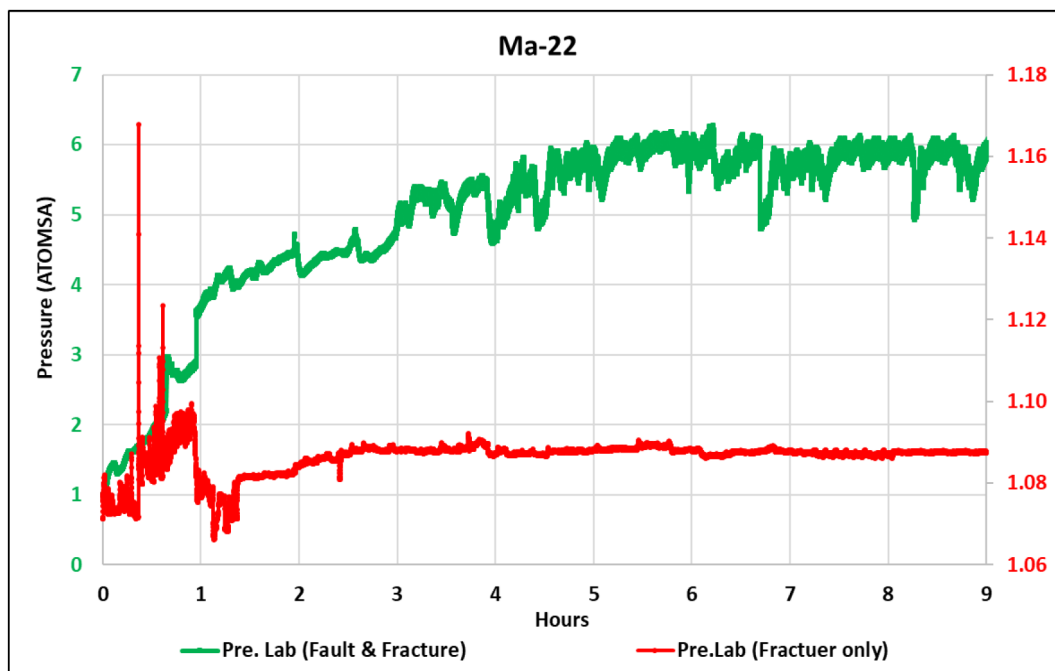


Figure 13—Pressure vs time from the tests for sample Ma-22. The red line is the upstream pressure during the water-flooding test on the fracture-only core plug. The green line shows the upstream pressure during the water-flooding test on the core plug with a fracture and partially sealing fault.

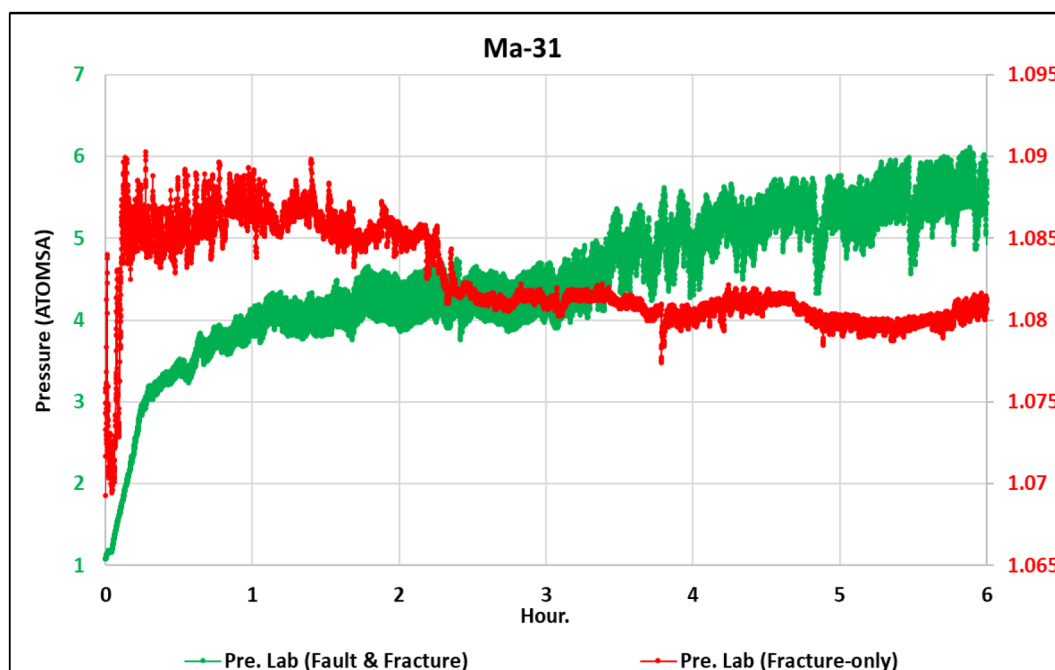


Figure 14—Pressure vs time from the tests for sample Ma-31. The red line is the upstream pressure during the water-flooding test on the fracture-only core plug. The green line shows the upstream pressure during the water-flooding test on the core plug with a fracture and partially sealing fault.

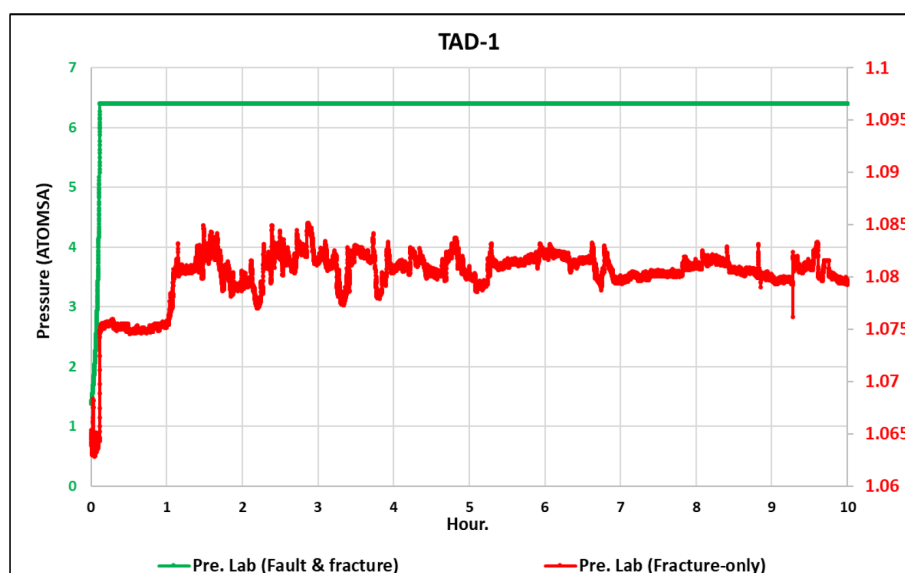


Figure 15—Pressure vs time from different tests for sample TAD-1. The red line is the upstream pressure during the water-flooding test on the fracture-only core plug. The green line shows the upstream pressure during the water-flooding test on the core plug with a fracture and partially sealing fault.

Saturation distribution using computed tomography

Table 5 presents the results obtained from the analysis of the oil saturation in sample Ma-22. The oil saturation in the sample was around 75% before water flooding and 55% and 66% after water flooding in the fracture-only plug and the core plug with a fracture and partially sealing fault, respectively. Computed tomography saturation measurements were taken on the upper and lower parts of the core plugs for comparison of the water saturation before and after the water-flooding tests (see Figure 16 and Figure 17). The results of the CT saturation measurement in the sample before water flooding showed that the upper part had a higher oil saturation (around 77%) than the lower part (around 73%). The oil saturation after

water flooding in the upper part was around 56% in the fracture-only core plug and around 65% in the core plug with a fracture and partially sealing fault. The oil saturation after water flooding in the lower part of the fracture-only core plug was around 53% compared to 66% in the lower part of the core plug with a fracture and partially sealing fault.

Table 5—Oil saturation before and after the water flooding results from the water-flooding test and computed tomography oil saturation measurement for sample Ma-22. Note that the upper part of the core produced more oil than the lower part in the core plug with the partially sealing fault.

Ma-22		So% before water flooding	So% after water flooding- (fracture only)	So% after water flooding (fracture and partially sealing fault)	Oil prod.% in (fracture only)	Oil prod.% in (fracture and partially sealing fault)
Water-flooding test		75	55	66	20	9
Avg. CT saturation measurement	Upper part	77	56	65	21	12
	Lower part	73	53	66	20	7
Average		75.0	54.5	65.5	20.5	9.5

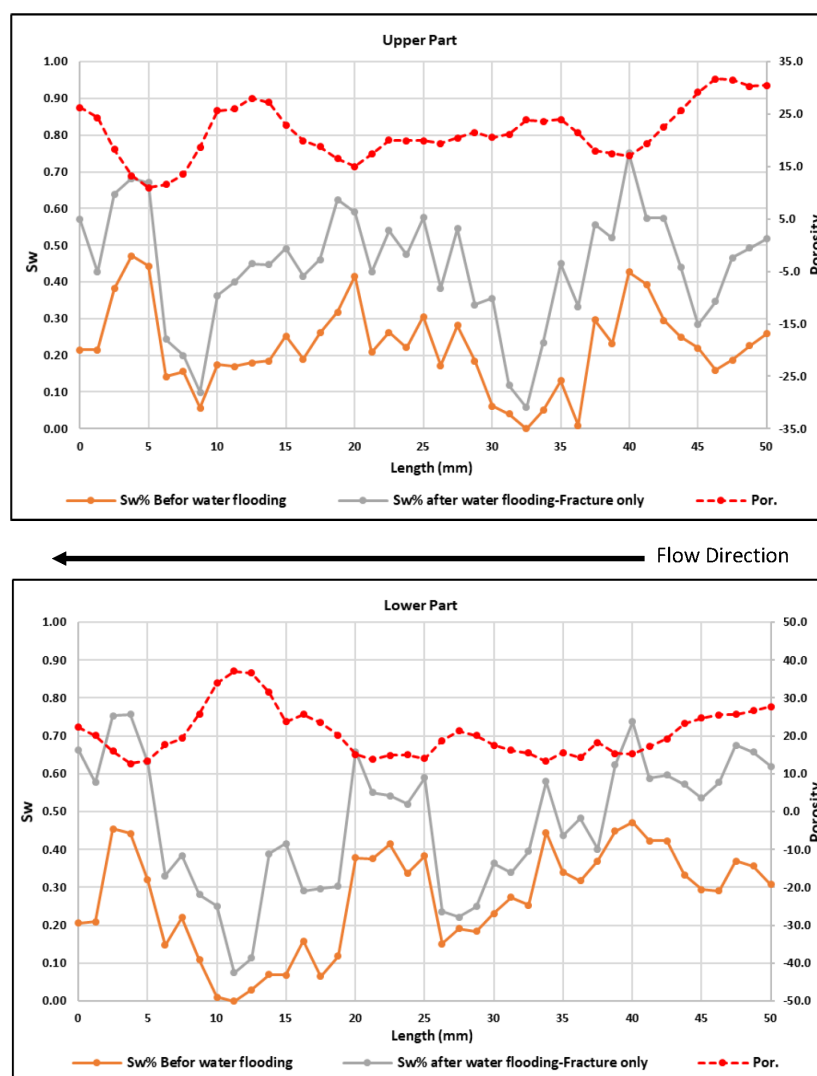


Figure 16—Water saturations and porosity variations along the core plugs from Ma-22 determined using computed tomography. Results are shown for the fractured core plug before and after water flooding.

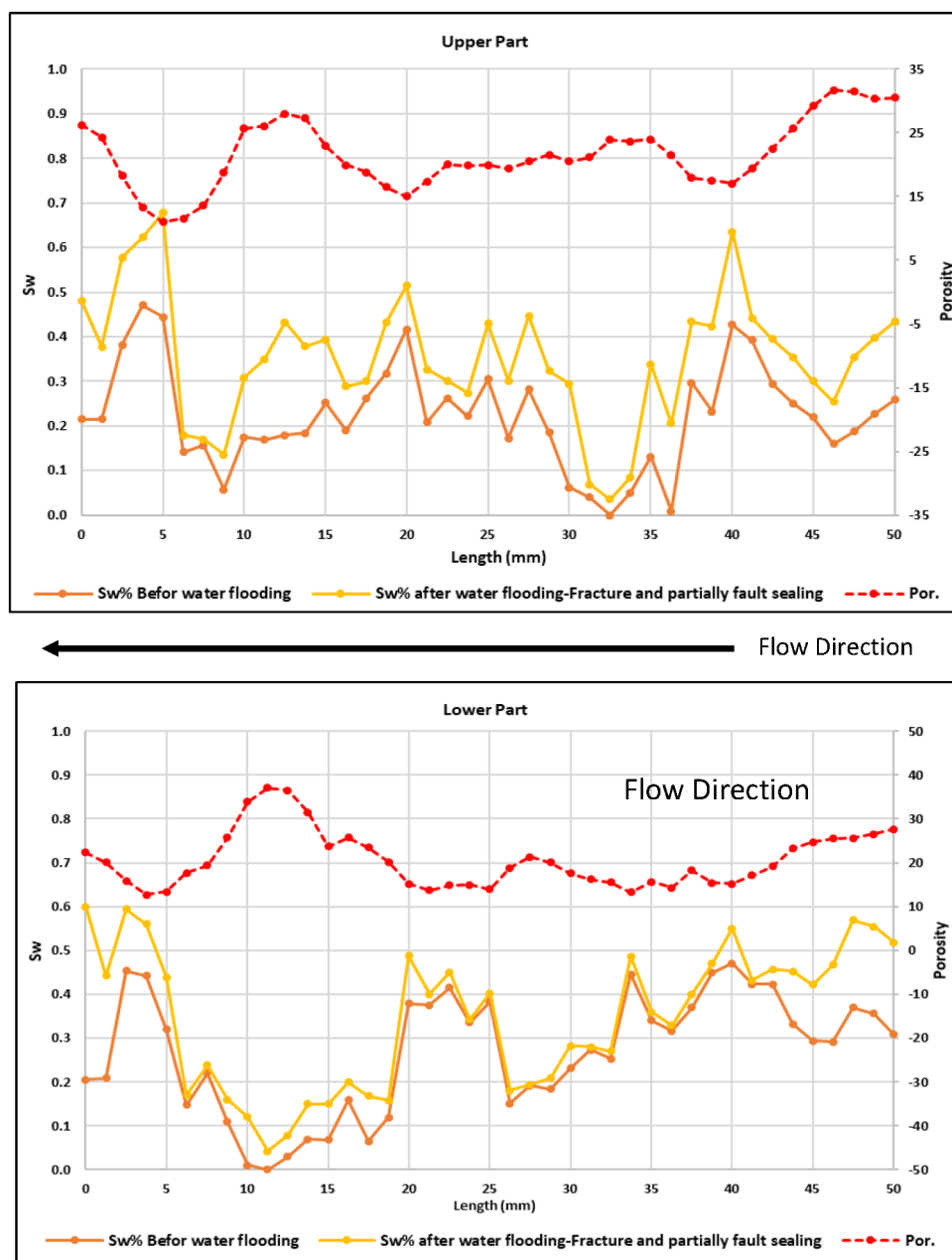


Figure 17—Water saturations and porosity variations along the core plugs from Ma-22 determined using computed tomography. Results are shown for the fractured core plug before water flooding and the core plug containing a fracture and a partially sealing fault after water flooding.

Table 6 provides the oil saturations measured in the Ma-31 core plug. The oil saturation in the sample was around 64% before water flooding and 49% after water flooding in the fracture-only core plug and 57% after water flooding in the core plug with the partially sealing fault. Saturation measurements were made using CT for the upper, middle and lower parts of the core plugs before and after the water-flooding tests. The results (see Figure 18 and Figure 19) show that the upper part has the highest oil saturation (around 66%) when compared with the middle and lower parts (63% and 62%, respectively). These differences could be caused by measurement errors or reflect the heterogeneous nature of the sample.

Table 6—Oil saturation results before and after water-flooding from the water-flooding test and computed tomography oil saturation measurement for sample Ma-31. Note that the highest production came from the lower part of the core, the lowest production from the middle part and intermediate production from the upper part.

Ma-31		So% before water flooding	So% after water flooding-(fracture only)	So% after water flooding (fracture and partially sealing fault)	Oil prod.% in (fracture only)	Oil prod.% in (fracture and partially sealing fault)
Water-flooding test		64	49	57	15	7
AV. CT saturation measurement	Upper part	66	52	59	14	7
	Middle part	63	48	57	15	6
	Lower part	62	45	53	17	9
Average		63.7	48.3	56.3	15	7

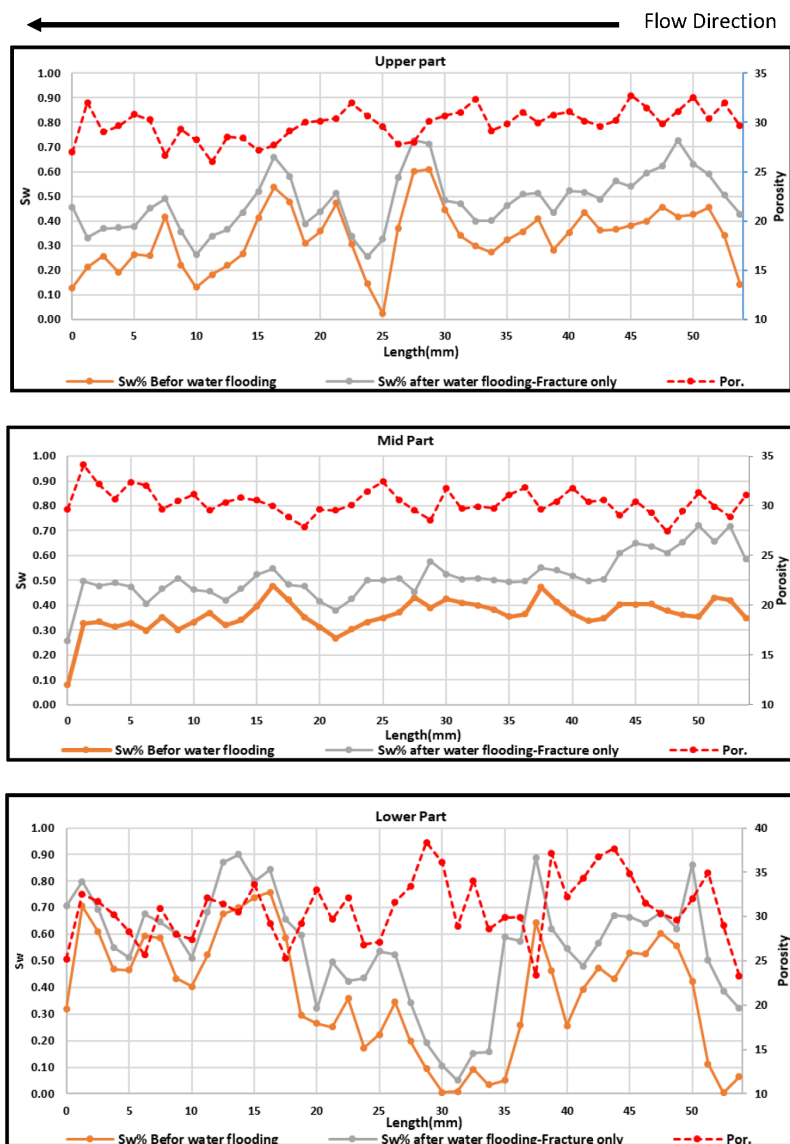


Figure 18—Water saturations and porosity variations along the core plugs from Ma-31 determined using computed tomography. Results are shown for the fractured core plug before and after water flooding.

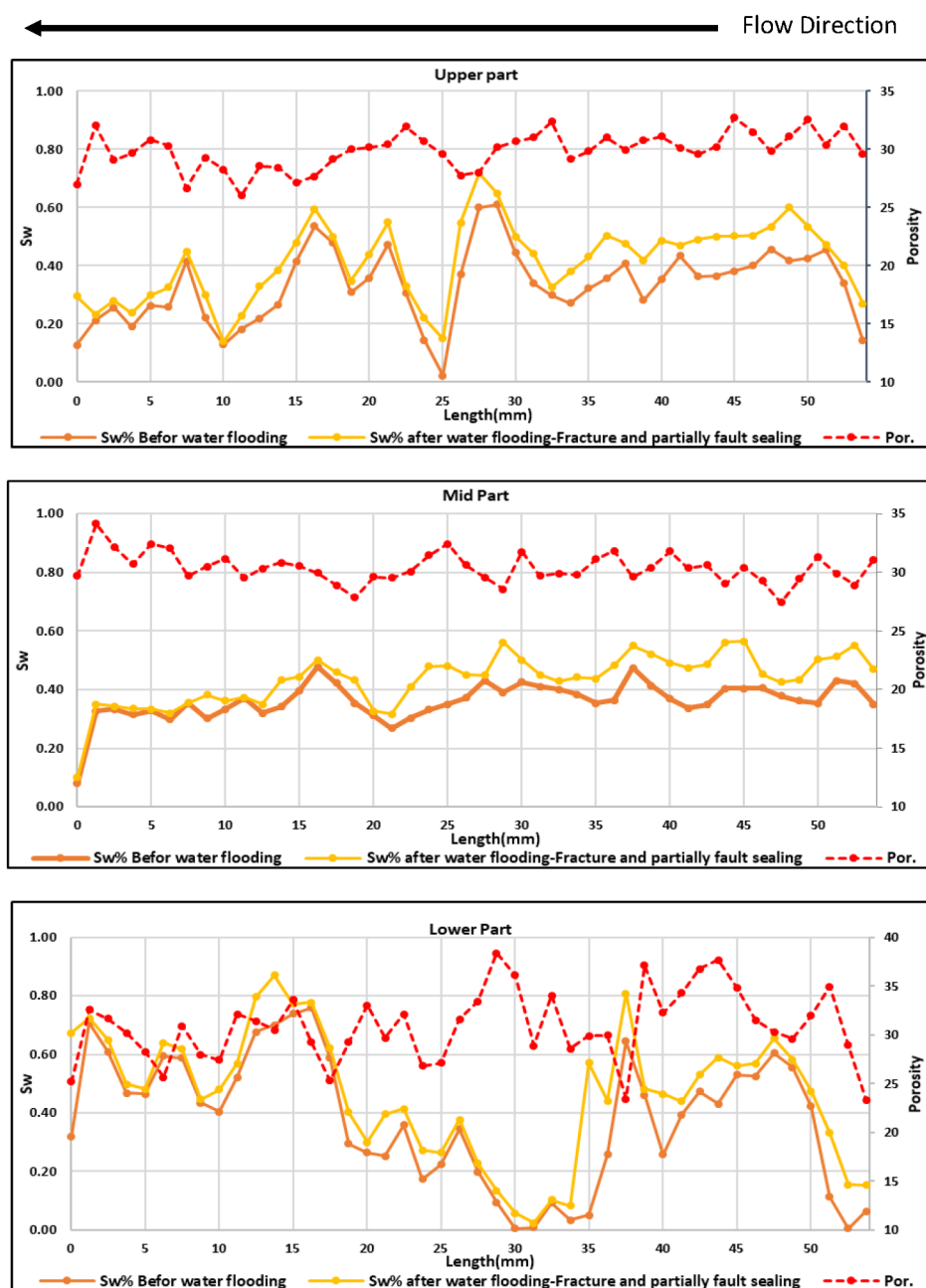


Figure 19—Water saturations and porosity variations along the core plugs from Ma-31 determined using computed tomography. Results are shown for the fracture-only core plug before water flooding and the core plug containing a fracture and a partially sealing fault after water flooding.

Oil saturation after water flooding was around 52% in the upper part of the fracture-only core plug and around 59% in the core plug with the partially sealing fault. The oil saturation after water flooding in the middle part of the core plug of the fracture-only core was around 48% compared to 57% for the sample with the partially sealing fault. The oil saturation after water flooding in the lower part of the fracture-only core plug was around 45% compared to 53% in the sample with the partially sealing fault.

Table 7 provides the oil saturations measured in sample TAD-1. The oil saturation in the sample was around 92% before water flooding and 86% after water flooding in the fracture-only core plug and 88% after water flooding in the core plug with the partially sealing fault. Computed tomography saturation measurements were taken on the upper and lower parts of the core plugs for comparison of the oil saturation in the core plug before and after the water-flooding tests (see Figure 20 and Figure 21). The results of

the CT saturation measurement in the sample before water flooding showed that the upper part and lower part had similar oil saturation (around 90%). The oil saturation after water flooding in the upper part was around 86% in the fracture-only core plug and around 84% in the core plug with a partially sealing fault. Correspondingly, the oil saturation after water flooding in the lower part of the fracture-only core plug was around 82% compared to 88% in the lower part of the core plug with the partially sealing fault.

Table 7—Oil saturations before and after water flooding for sample TAD-1. Note that the upper section of the sample produced more oil than the lower section.

TAD-1		So% before water flooding	So% after water flooding (fracture only)	So% after water flooding (fracture and partially sealing fault)	Oil prod.% (fracture only)	Oil prod.% (fracture and partially sealing fault)
Water-flooding test		92	86	88	6	4
Av. CT saturation measurement	Upper part	90	86	84	4	6
	Lower part	90	82	88	8	2
Average		90	84	86	6	4

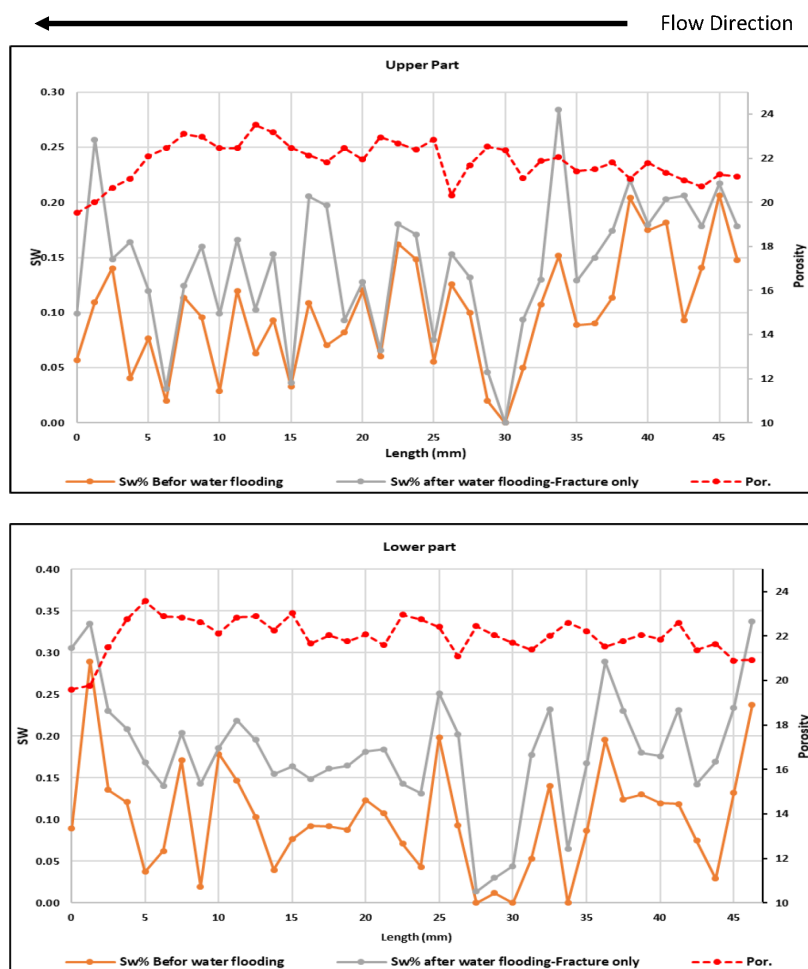


Figure 20—Water saturations and porosity variations along the core plugs from TAD-1 determined using computed tomography. Results are shown for the fracture-only core plug before and after water flooding

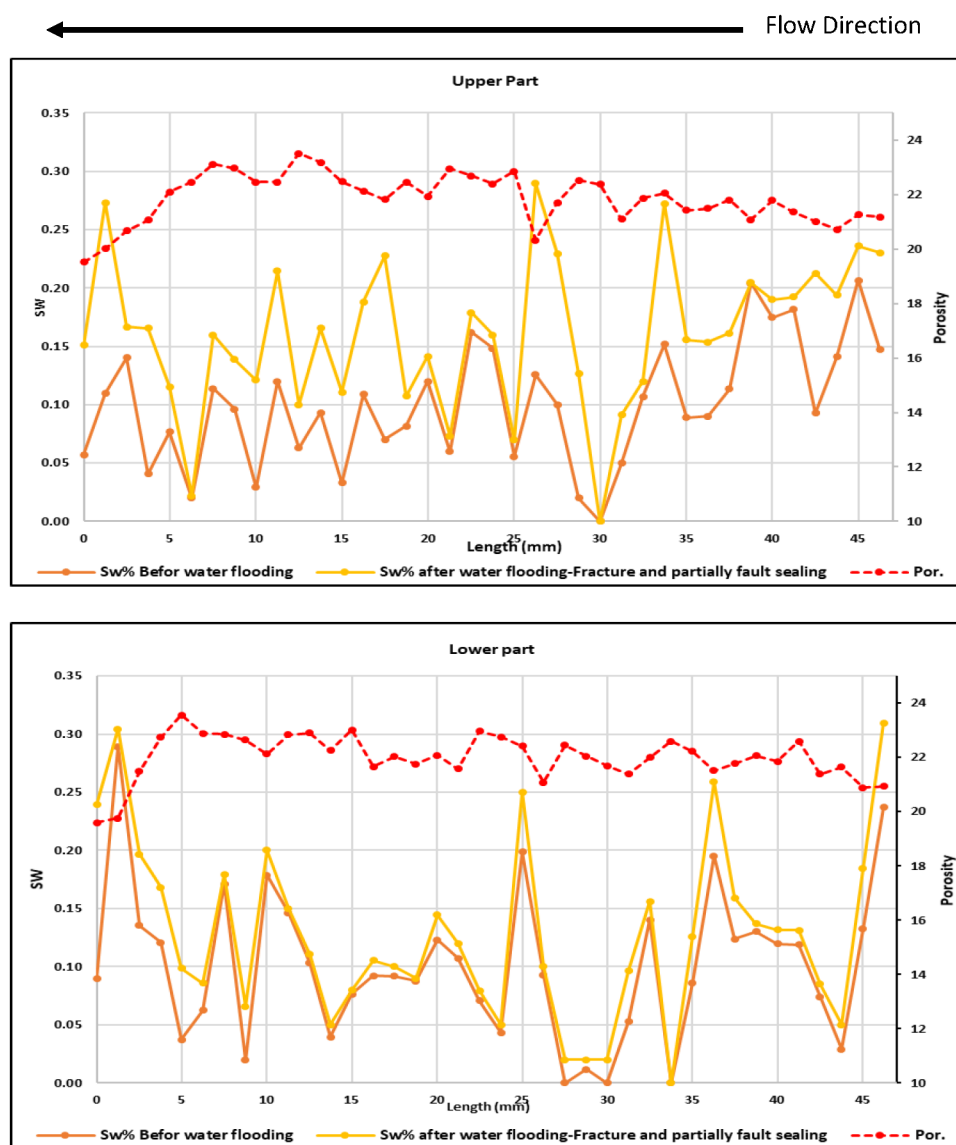


Figure 21—Water saturations and porosity variations along the core plugs from Ma-TAD determined using computed tomography. Results are shown for the fracture-only core plug before water flooding and the core plug containing a fracture and partially sealing fault after water flooding.

Discussion

Impact of the partially sealing fault on oil production

A key knowledge gap addressed in this thesis relates to the impact of low-permeability faults in conjunction with open fractures on oil production. To address this issue, six fluid-flow experiments were conducted on three carbonate core samples (Ma-22, Ma-31 and TAD-1).

The oil production from the two water-flooding tests conducted in sample Ma-22 is shown in Figure 22. Oil production from the fracture-only sample was 21% and 20% for the upper and lower parts of the core plug, respectively. Oil production in the partially sealing fault and fracture sample was 12% and 7% in the upper and lower parts, respectively. There are three potential reasons why the upper part of the sample with the partially sealing fault produced more oil than the lower part. First, the location of the leak point of the partially sealing fault meant that fluid was focused through the upper part of the sample. Second, the fault and fracture reduced oil production in the lower part due to the early water breakthrough in the upper part.

Finally, while the average porosity of the upper and lower parts (21.6% and 20%, respectively) are within an acceptable margin of error, other properties (e.g. permeability and pore size) could be slightly different.

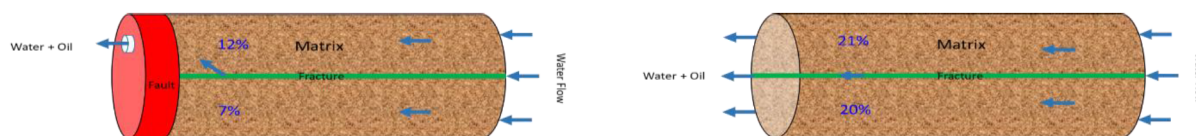


Figure 22—The oil production percentage in the upper and lower parts of the fractured core plug (right) and the partially sealing fault and fracture core plug (left).

A similar displacement pattern was observed in sample Ma-31 (see Figure 23). In particular, in the partially sealing fault and fracture core, oil production from the fracture-only sample was 14%, 15% and 17% for the upper, middle and lower parts, respectively. Oil production was 7%, 6% and 9% in the upper, middle and lower parts, respectively. The differences between the oil saturation in different parts of the core are within experimental error. However, slight differences could be real and have several possible explanations for why more oil was produced by the upper and lower parts based on the section size in the partially sealing fault and fracture sample. First, the location of the leak point focused brine through the matrix towards the leak in the middle section of the core. Second, the presence of two fractures meant that the upper and lower parts of the core plug were bypassed, reducing oil production from the middle part.

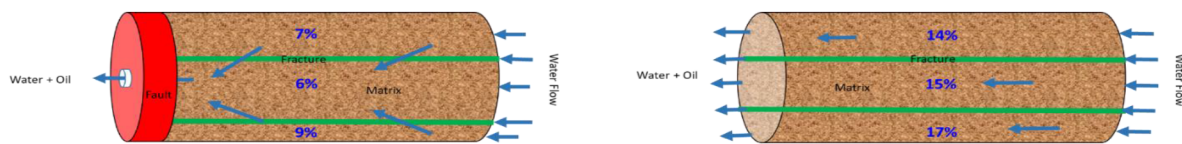


Figure 23—Oil production percentage in the upper and lower parts in the fracture-only core plug (right) and the partially sealing fault and fracture core plug (left).

In sample TAD-1, oil production from the lower part of the fracture-only core plug was 8% compared to 4% in the upper section (see Figure 24). The partially sealing fault and fracture sample exhibited the opposite behaviour, in which 6% of the oil was produced in the upper section and 2% in the lower section. Several factors could explain this observation. First, the location of the leak through the partially sealing fault was adjacent to the upper section, which meant that water was focused through this section. Second, the short distance between the fracture and the leak point of the partially sealing fault reduced oil production in the lower part due to the early water breakthrough in the upper part.



Figure 24—Oil production percentage in the upper and lower parts in the fracture-only core plug (right) and the partially sealing fault and fracture core plug (left).

In all samples, there was some oil production from the outlet during water flooding, but it could take half an hour to an hour for the oil to be produced, even if the injection rate and pressure remained constant or there was a slight change in the pressure reading of the core sample (see Figure 10 to Figure 15). The main reason for this is that brine has a significantly lower viscosity than oil, which results in some mobile and trapped oil being pushed out of the core.

The oil production from the Amott cell is compared to that of the water-flooding tests in Table 8. Around 95.7%, 83.3% and 77.8% of the oil was produced from the fracture-only samples Ma-22, Ma-31 and TAD-1,

respectively (i.e. an average of 85.6 %). Around 41.3%, 40% and 50% of the oil was produced from the samples of Ma-22, Ma-31 and TAD-1, respectively, in the fracture and partially sealing fault samples with average oil production of 43.8%. It is noteworthy that the sample that produced the least oil was Ma-31, which was affected by early water breakthrough from two fractures, while the other two samples, Ma-22 and TAD-1, only had one fracture.

Table 8—Test results and comparison between the average oil produced from the water-flooding tests in the fracture-only core plug and the fracture and partially sealing fault core plug.

Sample/test	Ma-22 (one fracture)	Ma-31 (two fractures)	TAD-1 (one fracture)	Avg. oil production
Oil production % from the water-flooding test (fracture only)	95.7	83.3	77.8	85.6
Oil production % from the water-flooding test (fracture with partially sealing fault)	41.3	40.0	50.0	43.8
Avg. oil produced in partially sealing fault over fracture-only samples				51.2

By the end of the laboratory experiments, the three samples with the partially sealing fault had produced an average of 51.2% of the total oil, which is half of the amount produced by the fracture-only samples. This indicates that the existence of a partially sealing fault in the core plug has a significant effect on total oil recovery. Furthermore, this provides evidence of an effect on oil production if partially sealing faults are found in a carbonate reservoir with open fractures and low-permeability faults.

Impact of faults and fractures on pressure.

Figure 13 to Figure 15 illustrate the injection pressure profile vs time from different tests for the three samples. It took several hours to achieve a constant pressure when injecting at 0.4 cm³/min. For example, the test on plug Ma-22 took around nine hours to achieve constant pressure from the start of water injection. This likely reflects the fact that two phases are moving at the start of the experiment, but it is mainly water being expelled towards the end of the experiment, and the pressure stabilises and remains constant in the last hours. Therefore, the experiment ended, indicating that the system had achieved a steady-state condition during the experiment, and there was no more oil production.

The pressures were far higher in the core plugs containing the partially sealing fault because the only outlet for the produced fluid was the small hole in the porous plate positioned next to the matrix to avoid direct production from the fracture. Unstable pressure in the sample with the partially sealing fault was due to fluid flow transported over considerable distances from the matrix to the open fractures and then through the partially sealing fault. For example, the pressure in sample Ma-31 (fracture only) (see Figure 13) declined and became stable 2.5 hours after starting the core-flooding experiment. On the other hand, the pressure increased in the core plug with a fracture and partially sealing fault. It could be that the two fractures in the core plug may have directly influenced pressure behaviour when they reached a steady-state condition with only one fluid flow in the system.

The impact of rock heterogeneity on water saturation

Computed tomography scanning was conducted at intervals of about 1.5 mm along the core length, giving different cross-sectional images covering the entire length of each core plug. Computed tomography scanning of the dry and water-saturated core plugs showed a heterogeneous porosity distribution. For example, Ma-22 had a porosity distribution ranging in the upper part from 11% to 32% and in the lower part from 13% to 37%. Ma-31 had a porosity ranging in the upper part from 26% to 33%, in the middle part from 27% to 34% and in the lower part from 23% to 38%. Sample TAD-1 had less heterogeneity, with

porosity in the upper and the lower parts having similar values ranging from 20 to 24%. It is possible that the injected water tended to first move to the smaller pores, which are typically found in the regions with lower porosity, with higher capillary entry pressure (see Figure 25). However, there is no direct evidence because the pores are too small to image with the CT scanner.

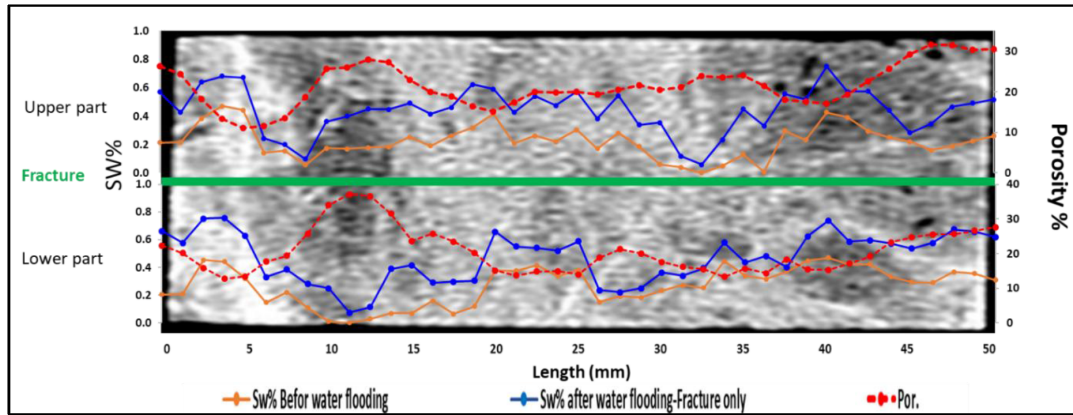


Figure 25—Example of a computed tomography (CT) scan of sample Ma-22 showing a CT porosity ranging in the upper part from 11% to 35% and in the lower part from 12% to 40%. Furthermore, Sw is highest in the parts of the sample with lower porosity both before and after water flooding.

Water saturation in the core plugs with partially sealing faults was less than in fracture-only core plugs (see Figure 26). This may be because the sealing fault acts as a barrier that reduces production, and the leakage point allows early water breakthrough from the fracture. In addition, water breakthrough affected the water saturation at the edge of the core plugs (outlet). In both cases, the water-flooding test in the partially sealing fault and fracture core plugs and in the fracture-only core plugs had a higher Sw (around 65%) than in the core plugs before the waterflooding test (around 20%). Overall, this suggests that the water was focused on the leak point and bypassed much of the oil in the samples.

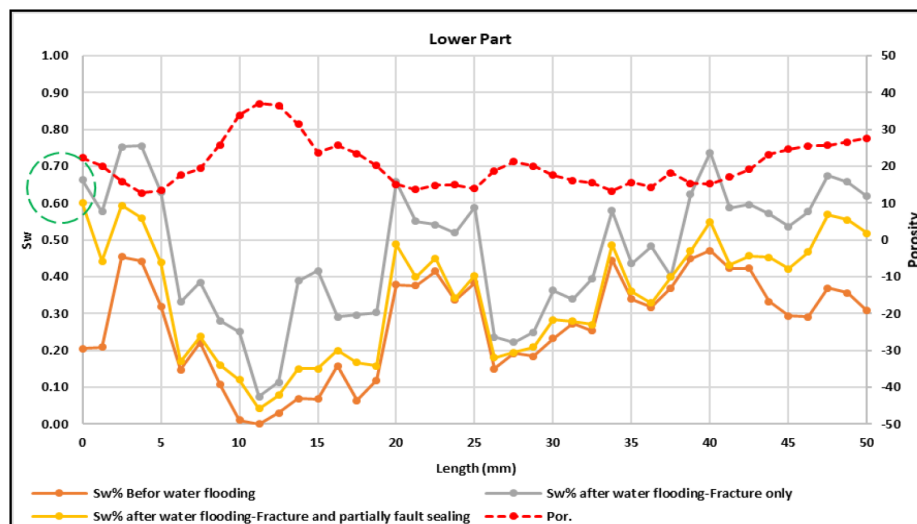


Figure 26—Water saturation and porosity distribution obtained using computed tomography for the upper part of the Ma-22 core plug. Results are shown for the fracture-only core plug before water flooding and the fracture-only core plug and the fracture and partially sealing fault core plug after water flooding. The green circle represents the high Sw at the edge of the core plugs (outlet) after water-flooding tests.

Conclusion

This paper has presented the results from an experimental study to assess how a partially sealing fault would impact oil production from a reservoir in which open fractures were present. The experiments found that oil production from the fracture-only samples was 86% compared to 44% from the partially sealing fault and fracture samples. This indicates that the existence of a partially sealing fault in the core plug has a significant effect on the total oil recovery. Several factors play a significant role in the impact of fault-fracture interaction on fluid flow in carbonate rock:

- The location of the leak point of the partially sealing fault. In particular, more oil was produced from the section of the plug that was juxtaposed against the leak point.
- The location of the partially sealing fault. The fracture worked to reduce oil production from the lower part due to the early water breakthrough when the partially sealing fault was next to the upper part of the sample.
- The location of the leak. This had an impact on the oil distribution at the end of the test. In particular, water was focused through the leak point, and more oil was produced from the section adjacent to the leak point.
- The distance between the fracture and the leak point. The pressure throughout the water-flooding tests depended on the location of the leak point and the distance to the fracture in the partially sealing fault samples rather than the matrix permeability.

The DMM was used to accurately measure the oil flow rate during water-flooding experiments. The method involved measuring the resistivity of the fluid being produced by the core and then classifying the fluids based on this value.

The pressures recorded during the first few hours of the core-flooding experiments were variable, probably because the ratio of oil to water flowing in the sample varied. However, the pressure became constant when only water was flowing later in the experiment. Furthermore, unstable pressure in the sample containing the partially sealing fault was due to fluid flow transported over considerable distances from the matrix to the open fractures and then through the partially sealing fault being affected by differences in relative permeability; this could lead to higher pressure and a decrease in oil production.

The images obtained using X-ray CT showed that the core plugs had a heterogeneous porosity distribution. It is possible that the injected water tended to move into the smaller pores first, which are typically found in regions with lower porosity and higher capillary entry pressure. In addition, water saturation in the core plugs with the partially sealing fault was lower than in fracture-only core plugs due to the fault sealing area acting as a barrier that reduced production, and water was focused on the leakage point by early water breakthrough from the fracture.

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